Chinook Salmon WY2024-25: Individual assignments with RoSA Integration

Genidaqs lab of Cramer Fish Sciences

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Assignment details

Genotypes are composed of new Chinook Salmon panel (Anderson et al. 2025; https://doi.org/10.1111/eva.70110). The bioinformatics uses the **snakemake pipeline** and **microhaplotopia**.Genetic stock identification is conducted using **rubias** and the associated NMFS reference genetic baseline (Anderson et al. 2025).

- Mega-Simple-Microhap-Snakeflow
- Microhaplotopia
- Rubias

Total number of samples analyzed to date

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## Total Number of Samples Analyzed to Date: 1397
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New samples added since last report

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## Number of New Samples Added Since Last Report: 9
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Assignment summary for new samples

Table 1: Population assignments for newly added samples

Facility	Genetic_Assignment	Spring_Pop_Assignment	count
CVP	Fall	NA	8
SWP	Fall	NA	1

Cumulative population assignment summary

Table 2: Cumulative genetic assignments by location (Facility)

Facility	Genetic_Assignment	Spring_Pop_Assignment	count
CVP	Fall	NA	989
CVP	Late Fall	NA	8
CVP	Spring	Feather River	34
CVP	Spring	Mill/Deer Creek Spring	1
CVP	Winter	NA	11
SWP	Fall	NA	331
SWP	Late Fall	NA	1
SWP	Spring	Feather River	22

Table 3: Cumulative length-at-date model classifications by location (Facility)

Facility	LAD_Race	count
CVP	Fall	778
CVP	Late Fall	10
CVP	Spring	226
CVP	Winter	29
SWP	Fall	231
SWP	Spring	117
SWP	Winter	6

Appendix: Assignment data file description

There is an assignment data file associated with this report. Below is a description of the fields present in the .csv file.

CVTA_ID -------------- Individual identifier (e.g., CVTA ID, field ID)
SampleDateTime ------- Sample data and time individual was collected (yyyy-mm-dd h:mm)
ForkLength ----------- The fork length of individual

Prob_Assignment ------ Probability of Assignment. The probability that an individual belongs to a specific reporting unit (repunit), calculated from the model's posterior distributions. Range: 0 to 1, where a higher value (closer to 1) indicates stronger evidence for assignment to that repunit.
rosa_classification -- Label for interpreted rosa_genotypes_str
Genetic_Assignment --- Final assignment considers both reporting unit and rosa_classification. Final assignment currently is created using the following criteria, although this will be monitored given observed results and Agency input.

- Individuals assigning to Fall with an Intermediate or Late RoSA haplotype will be classified as Fall.
- Individuals assigning to Fall with an Early RoSA haplotype will be classified as Spring.
- Individuals assigning to Fall and ColemanLF collection with an Intermediate or Late RoSA haplotype will be classified as Late Fall.
- Individuals assigning to Spring with a Late RoSA haplotype will be classified to Fall.
- Individuals that assign to Winter will retain the Winter reporting unit irrespective of RoSA haplotype.

Spring_Pop_Assignment - If applicable

- Individuals assigning to Fall with an Early RoSA haplotype will be classified as Feather River.
- Individuals assigning to Spring with an Early or Intermediate RoSA haplotype will be classified to collection (Butte Creek Spring, Mill/Deer Creek Spring).

Facility ----------- Location identifier
LAD_Race ----------- Length-at-date model classification (e.g., Delta Model)
Date_Assigned ------ Date assignment was added to the data file and report (%Y-%m-%d)

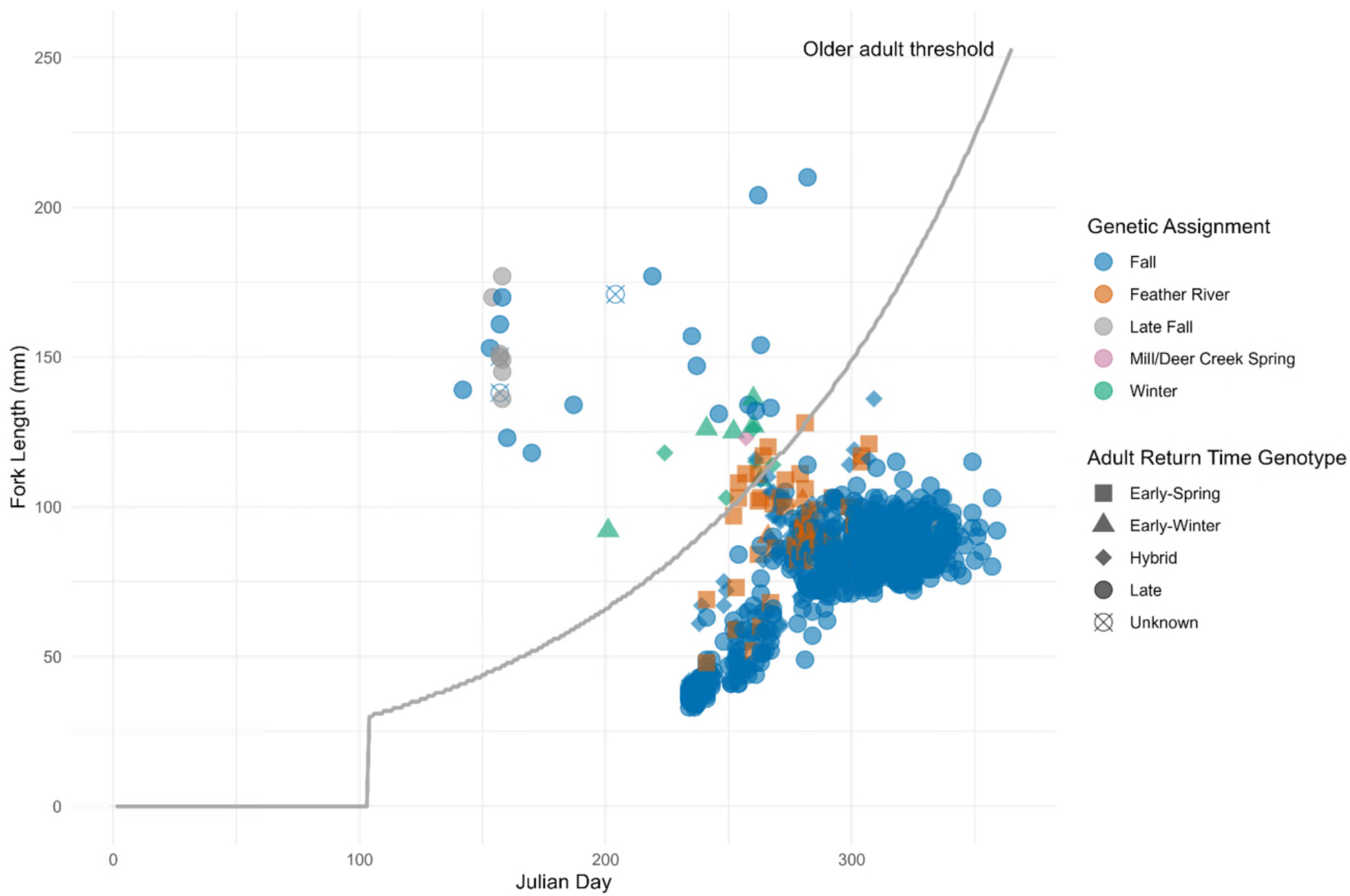

Figure. Genetic assignments of Chinook Salmon captured at salvage facilities 2024-25 by Julian Day (July 1st – June 30th) and Fork Length, with associated adult return time genotypes (i.e., RoSA).

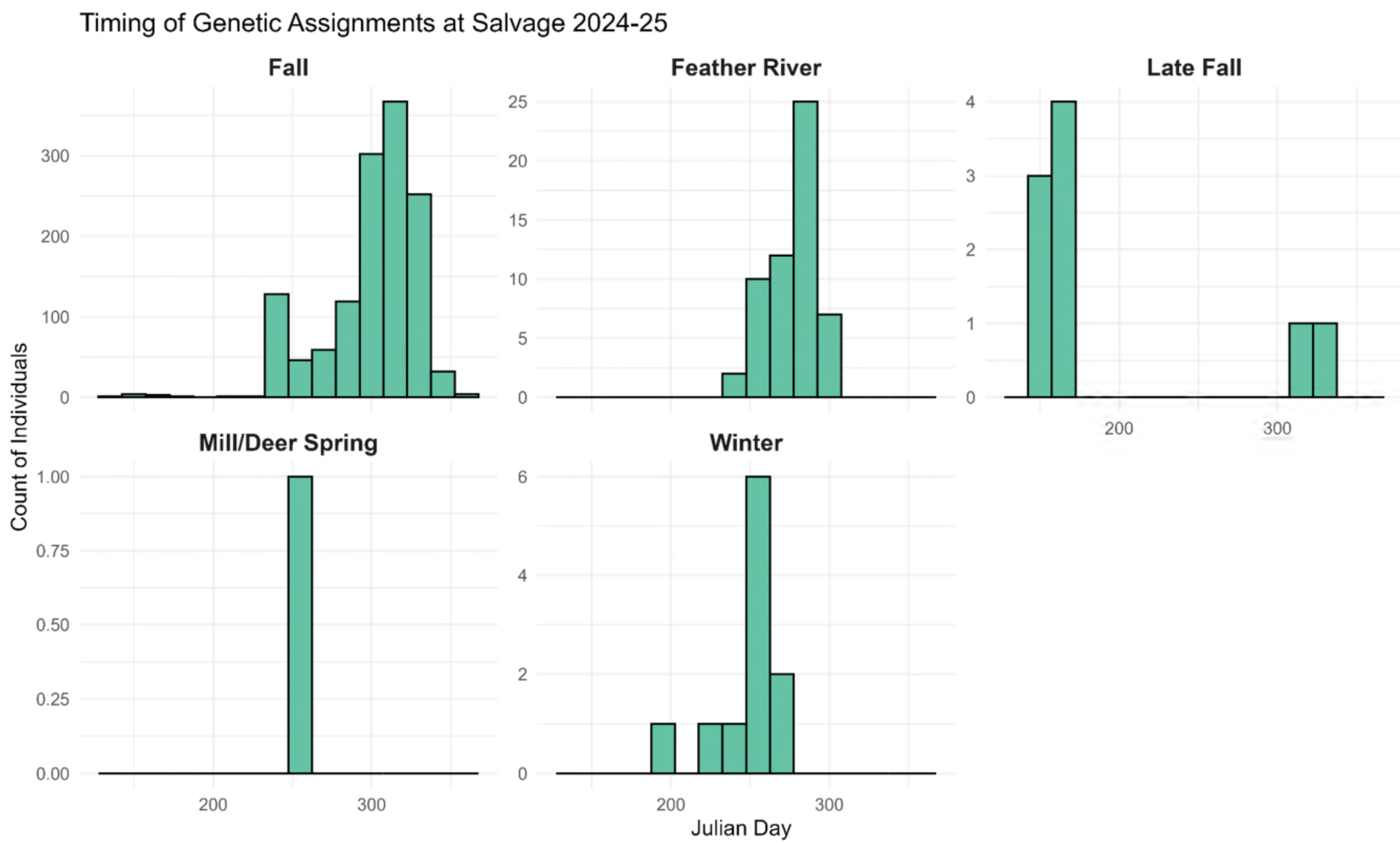

Figure. Genetic assignment counts of Chinook Salmon captured at salvage facilities 2024-25 by Julian Day (July 1st – June 30th).